

Interoperable digital product passport and logbook platforms for circular construction materials, enabling extended producer responsibility compliance and lifecycle traceability.

Construction × Product traceability and digital product passport × Cloud data platform · OS091 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
60 is the world moving	34 can we play, can we win	HIGH 0 named in the evidence	€1.8bn observed evidence · per year	LATER research and standards activi...	L2 14 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 1.0 published reference(s) in Construction, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity involves building interoperable digital product passport and logbook platforms for circular construction materials, enabling extended producer responsibility compliance and lifecycle traceability. It targets construction material manufacturers, contractors, and building owners who need to comply with emerging EU regulations and demonstrate circularity. The need is driven by regulatory mandates and industry research, though market adoption is still in early stages.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€4.1bn €301m – €16.2bn	€1.8bn €135m – €7.3bn	€4.6m €337k – €18.2m	observed
Observed: tendered contract value, annualised	€123m €123m – €471m	€123m €123m – €471m	€309k €309k – €1.2m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Construction (EU27_2020)	250,045 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Cloud data platform	18.9 % of enterprises (10+ employees)	observed	Eurostat · isoc_cicce_usen2 · E_CC_PDB	2025
Annual value of one engagement	€952k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	22,551 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

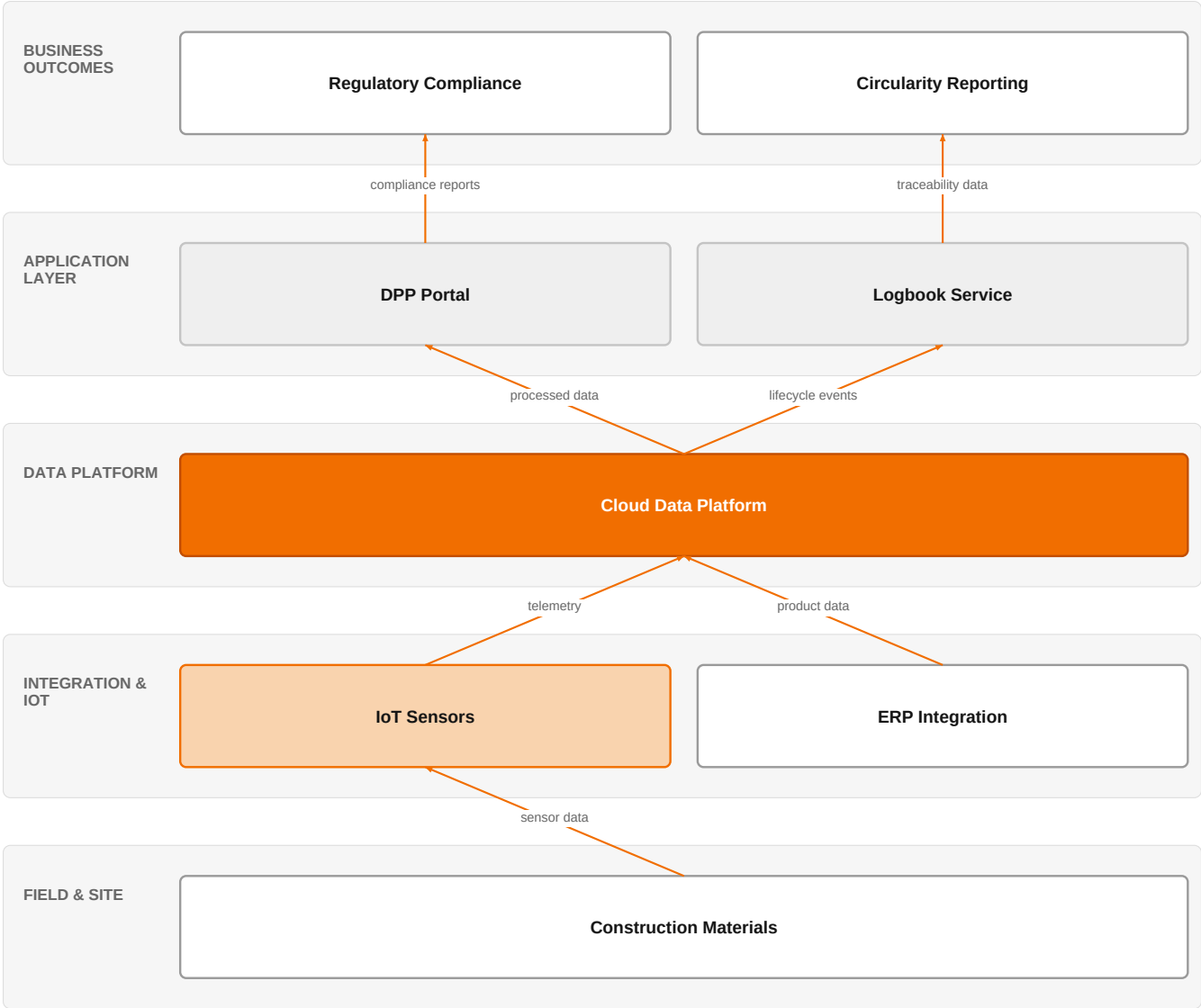
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€123m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07

Assumed obtainable share of the serviceable market	0.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a
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Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 24 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Digital Product Passport Platform for Construction



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's cloud data platform integrates IoT and ERP data to power DPP and logbook services, enabling regulatory compliance and circularity reporting.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU's Ecodesign for Sustainable Products Regulation (ESPR) mandates Digital Product Passports for electronics and appliances by 2027, creating an urgent need for machine-readable product lifecycle data [SIG-BA196FF05244]. Research highlights the construction sector's need for circularity, with the BUILDERS project addressing extended producer responsibility schemes [SIG-36E0C3234EE0]. A Digital Product Logbook is proposed as a repository for logging data after product placement, filling a gap in the DPP framework [SIG-8E9F663B741B]. However, significant knowledge gaps remain in the construction industry regarding DPP implementation [SIG-EFEBD520406D].

Sources: SIG-BA196FF05244 Springer Science and B 2026-02-03; SIG-36E0C3234EE0 cordis.europa.eu 2026-06-01; SIG-8E9F663B741B Informes de la Constru 2026-03-30; SIG-EFEBD520406D openalex.org 2025-10-15

Who buys, and why now

The buyer is typically a sustainability or compliance officer at a construction material manufacturer or a large contractor, while the pain is felt by operations and supply chain teams who struggle to track material provenance and lifecycle data. The trigger is often an upcoming regulatory deadline or a major tender requiring circularity documentation. They need to prove compliance with EPR schemes and provide lifecycle traceability to win projects and avoid penalties.

Why it is hot now — every claim bound to a dated source

- The BUILDERS project addresses circularity in construction through extended producer responsibility schemes, which DPPs support. [SIG-36E0C3234EE0]
- Research proposes a Digital Product Logbook as a repository for logging data after product placement, relevant for construction. [SIG-8E9F663B741B]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a cloud data platform, leveraging its AI/Data/Cloud experts and IoT experts to build a data ingestion and management layer. The platform would integrate with existing customer systems and partner cloud services (Microsoft, AWS, Google Cloud) to create a scalable infrastructure. The engagement would start with a discovery phase to map data sources, then design a data model aligned with DPP standards, and finally deploy a pilot with a select set of products.

Why Orange can win

Orange can win by combining its cloud data platform expertise with its IoT capabilities and research from Orange Group researchers. The ability to integrate with multiple hyperscalers (Microsoft, AWS, Google Cloud) provides flexibility. However, the assets are relatively generic; the differentiator would be Orange's ability to orchestrate a sovereign, secure data ecosystem, though this is not explicitly stated in the evidence.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Microsoft	partner	L2	partner provides the required technology	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
Cloud data platform	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

HIGH competitive intensity (score 8.6; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Cloud data platform; competes in OX: Smart Industries — also an Orange partner	structural	—
Google Cloud	Hyperscaler	sells Cloud data platform — also an Orange partner	structural	—
Microsoft	Hyperscaler	sells Cloud data platform — also an Orange partner	structural	—
Accenture	Global systems integrator	sells Cloud data platform; competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	sells Cloud data platform; competes in OX: Smart Industries	structural	—

DXC Technology	Global systems integrator	sells Cloud data platform	structural	—
Infosys	Global systems integrator	sells Cloud data platform	structural	—
Sopra Steria	Global systems integrator	sells Cloud data platform	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What regulatory or customer requirements are driving your need for product lifecycle traceability?
- 2 Do you currently have any system for tracking material provenance or lifecycle data?
- 3 Which construction products or materials are your priority for circularity compliance?
- 4 How do you currently share product data with downstream partners or regulators?
- 5 What is your timeline for implementing a digital product passport or logbook solution?

Objections you will meet

They will say	You can answer
We already have a cloud platform from AWS/Microsoft/Google.	That's true, and we can work with your existing cloud investment. Our value is in the data model and integration layer that makes DPPs interoperable across your supply chain, not in replacing your infrastructure.
DPP standards are still evolving; we don't want to invest prematurely.	You're right that standards are maturing. That's why we propose a flexible data architecture that can adapt to final regulations, and we can start with a pilot to prove value without a large upfront commitment.
We don't have the data quality or granularity needed for DPPs.	That's a common challenge. Our IoT experts can help you identify what data is feasible to capture and design pragmatic solutions to fill gaps, starting with the most critical data points.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Assemble a bid angle on Product traceability and digital product passport with Cloud data platform and Microsoft (L2).
Sales	Open with Construction accounts on Product traceability and digital product passport — confirm an internal proof point before the meeting.
Strategist / Innovator	Commission a deep-dive on Product traceability and digital product passport in Construction using Cloud data platform: size the opportunity bottom-up and decide whether to prototype next quarter.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-36E0C3234EE0	2026-06-01	cordis.europa.eu	1	Building Up Industry Loops through Development, evaluation, and testing of an Extended producer...
SIG-63476A345BB2	2026-06-01	cordis.europa.eu	1	Fostering Resilient and Auto-Compliant Manufacturing Value Chains through Advanced Digital Prod...
SIG-36FFB2435F8A	2026-04-20	openalex.org	1	A Comprehensive Review of Digital Product Passports in Sustainable Manufacturing: Current State...
SIG-8E9F663B741B	2026-03-30	Informes de la Construcci...	1	Digital Product Logbook, the missing piece in the European Digital Product Passport puzzle
SIG-F47C31AB60CB	2026-02-14	MDPI AG	1	Dynamic Traceability and Reliability Management: Integrating the Digital Product Passport with ...
SIG-BA196FF05244	2026-02-03	Springer Science and Busi...	1	Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...
SIG-CC28A63D0BB7	2026-01-01	openalex.org	1	From Compliance to Circularity: Stakeholder Perspectives on Adopting Digital Product Passports ...
SIG-D0815CA3A107	2026-01-01	openalex.org	1	Representing Executable Circular Economy R-Strategies using Behavior Trees Embedded in Digital ...
SIG-30E389C313E3	2025-12-01	openalex.org	1	Integrating Digital Product Passports and BIM for Circular Construction: A Review of Challenges...
SIG-EFEBD520406D	2025-10-15	openalex.org	1	Stakeholder perspectives on digital product passports for construction products
SIG-40DC35D6534F	2025-10-14	openalex.org	1	Predictive Generation of Digital Product Passport as Decision Support in Circular Networks
SIG-6FBECA0F7EA7	2025-09-10	openalex.org	1	Design of a Sensor-Based Digital Product Passport for Low-Tech Manufacturing: Traceability and ...
SIG-48D7CEFBFA9E	2026-07-20	Stockholm Environment Ins...	2	From policy to practice: challenges of environmental data generation and Digital Product Passpo...
SIG-BE9AC623C90C	2026-07-02	Cambridge University Pres...	2	Towards a reference architecture for digital product passports in engineering - Cambridge Unive...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); risks_and_unknowns (missing from the model output); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:54+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.